

MAGNETOM

Troubleshooting Guide System

Gradient System

Aera
Skyra
Avanto fit
Skyra fit

Document Version

Siemens Healthcare GmbH reserves the right to change its products and services at any time.

In addition, manuals are subject to change without notice. The hardcopy documents correspond to the version at the time of system delivery and/or printout. Versions to hardcopy documentation are not automatically distributed.

Please contact your local Siemens Healthineers office to order a current version or refer to our website <http://www.siemens-healthineers.com>.

Disclaimer

Siemens Healthcare GmbH provides this documentation “as is” without the assumption of any liability under any theory of law.

The content described herein shall be used by qualified personnel who are employed by Siemens Healthcare GmbH or one of its affiliates or who are otherwise authorized by Siemens Healthcare GmbH or its affiliates to use such documents.

Assemblers and other persons who are not employed by or otherwise directly affiliated with or authorized by Siemens Healthcare GmbH or one of its affiliates are not entitled to use this documentation without prior written authority.

Copyright

“© Siemens Healthineers, 2010” refers to the copyright of a Siemens Healthineers entity such as:

Siemens Healthcare GmbH - Germany

Siemens Shenzhen Magnetic Resonance Ltd. - China

Siemens Shanghai Medical Equipment Ltd. - China

Siemens Healthcare Private Ltd. - India

Siemens Medical Solutions USA Inc. - USA

Siemens Healthcare Diagnostics Inc. - USA and/or

Siemens Healthcare Diagnostics Products GmbH - Germany

1	General Information	4
1.1	Notes	4
1.1.1	Safety information and/or references to safety information	4
1.1.2	General Remarks	5
1.1.3	Product specific Remarks	6
1.2	Requirements	8
1.2.1	Additionally Required Documents	8
1.2.2	Required Aids and Tools	8
2	Procedures	9
2.1	General	9
2.2	GPA	10
2.2.1	D100 Rectifier - Verifying a defect	10
2.2.2	Troubleshooting GPA Current Sensors	11
2.2.3	GPA Problems related to Implant Imaging Suite Option (Aera with FPO)	13
2.3	Gradient Supervision (VLM)	14
2.3.1	General Troubleshooting VLM	14
2.3.2	Special Problem - Incorrect Air Flow	16
2.3.3	Special Problem - Incorrect VLM Configuration	23
2.3.4	System Inspection after VLM FIRE Alarm	25
2.3.5	Appendix - Diagnostic Software (Siemens Medical VSC)	34
3	Changes to Previous Version	43
4	List of Hazard IDs	44

1.1 Notes

1.1.1 Safety information and/or references to safety information

 Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad. (TDK equipment type 8005 from 3M Deutschland GmbH).

WARNING

Please follow the general safety guidelines (→ General Safety Notes / TD00-000.860.01) as well as the MR-specific safety information (→ MR-specific safety information / MR-000.860.01)!

Failure to comply with the safety notes may result in death or serious injury.

- » Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Adhere to the information provided at all times.

CAUTION

Permanent Mains voltage at Mains filter input and circuit breaker F1/GPA at lower connections.

There is the risk of electrical shock.

- » Switch off mains power of the site, secure against switching on and check the voltage at the Mains Input filter.

CAUTION

Switching-off circuit breaker F1/GPA will not remove power from the GSSU power supply.

Risk of electrical shock is present.

- » Also switch off F21/EPC (Aera/ Skyra) or F2/GSSU (Avanto^{fit}/ Skyra^{fit}) in the GPA, before start working at the GSSU power supply.

CAUTION

High voltage across capacitor/ gradient cable connections.

- » After turning off the cabinet, the voltage decreases to a harmless level after 20 minutes.

**WARNING**

Troubleshooting and repairs involving current sensors, gradient filters, gradient coil, gradient cables may lead to problems with the system's image orientation.

This could result in a misdiagnosis.

- » Perform the "Image Orientation Check" after performing all repair procedures and disconnecting/opening the gradient coil power circuit, e.g., replacement of:
- » - Current sensors
- » - Gradient filters
- » - Gradient coil
- » - Gradient cables

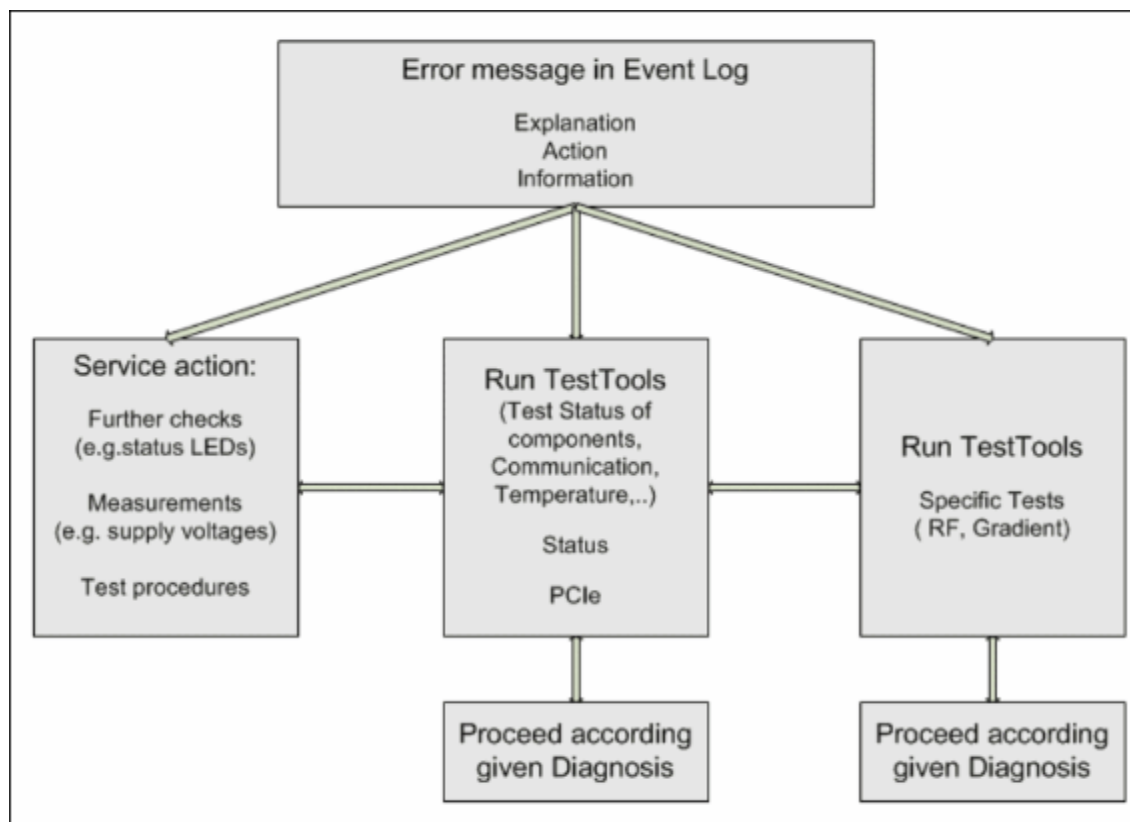
1.1.2 General Remarks

The main tool for troubleshooting consists of error messages delivered from the GPA which are shown in the error log file (Event Log) of the service platform. Click on the message IDs there to get further advice.

System component	Error message source (enter this string in the EventLog field "source" to filter for component-specific messages)
GPA K2309 (Aera/ Skyra)	MRI_GDV
GPA_XXL or GPA_XXL_F (Avanto ^{fit})	MRI_GAV
GPA K2298 (Skyra ^{fit})	MRI_GVE

1.1.2.1 General strategy for troubleshooting

Fig. 1: General proceeding in trouble-shooting



Further troubleshooting informations e.g. error codes can be found in the Service Knowledge Base (SKB).

1.1.3 Product specific Remarks

System	Aera 1.5 T		Skyra 3T	Avanto ^{fit}	Skyra ^{fit}
Grad. option	XJ-Gradient	XQ-Gradient	XQ-Gradient	SQ-Gradient	VQ-Gradient
GPA - Type	62/20/F	90/22/F	75/22/F	XXL_F	90/22
Material no.	10513280	10513350	10513313	10743860 or XXL 7564003	10046406

System	Aera 1.5 T		Skyra 3T	Avanto ^{fit}	Skyra ^{fit}
Gradient coil Material no.	10358395	10358260	10358500	08119690	10048321
Gradient coax cables Material no.	10358342	10358620	10358620	08119625	10048363

1.2 Requirements

1.2.1 Additionally Required Documents

Function Description "Gradient" (→ Interactive Functional Description / M7-000.850.01)

Replacement of Parts "Gradient" (→ Gradient System / M7-000.841.16)

Error messages Gradient

Wiring Diagrams, Jumper Lists "Gradient"

1.2.2 Required Aids and Tools

- Standard tool set
- DVM

2.1 General

In this section, you will find troubleshooting procedures for specific gradient problems.



Further troubleshooting informations e.g. error codes can be found in the Service Knowledge Base (SKB).

2.2 GPA

2.2.1 D100 Rectifier - Verifying a defect



WARNING

High voltage across capacitor/ gradient cable connections!

Risk of electrical shock and component damage! Contact with voltage conducting parts will lead to death or serious physical injury!

- » After turning off the cabinet, the voltage decreases to a safe level after 20 minutes.
- » Make sure, contactor K1+K2 is off when you plug in X31, X32 back into the final stage.
- » Do not plug in X31, X32 again when K1+K2 is on.
- » Adhere to the described steps for your own safety.

What:

- It's sometimes hard to identify a defective D100 rectifier PCB in the gradient cabinet.
- This information should help to verify the defect.

Where:

- GPA_XXL
- GPA_XXL_F
- GPA 75/22/F
- GPA 90/22
- GPA 90/22/F

When:

- During troubleshooting

Scenario:

- LEDs on D100 are OFF, circuit breaker(s) have tripped-off at arrival of the CSE.

Procedure:

- Switch-off circuit breaker F21/EPC (Aera/ Skyra) or F2/GPA (Avanto^{fit}/ Skyra^{fit}) and F1/GPA.
- Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.
- Disconnect power cables X31, X32 (all other cables remain connected) of the corresponding final stage.
- Turn-on the tripped-off circuit breaker.
- Switch-on circuit breaker F21/EPC (Aera/ Skyra) or F2/GPA (Avanto^{fit}/ Skyra^{fit}) and F1/GPA.
- Power-up the gradient cabinet and evaluate cases, see below.

- After case evaluation:
 - Switch-off circuit breaker F21/EPC (Aera/ Skyra) or F2/GPA (Avanto^{fit}/ Skyra^{fit}) and F1/GPA again.
 - Wait at least 20 minutes to ensure the final stage capacitors are fully discharged.
 - Reconnect power cables X31, X32.
 - Replace the defective component. See Replacement of Parts description.

Case:

- Cabinet starts-up and LEDs on D100 are glowing => D100 is OK, replacement not necessary.
- Circuit breaker(s) at the transformer trip-off again, or gradient cabinet does not start (overload during power-on: K1 is actuating but drops-off again before K2 switches on). => D100 rectifier is broken and needs to be replaced.
- Cabinet starts up, but LEDs on D100 remain OFF => supervision on D100 is defect or rectifier defect, replacement of D100 is necessary.

2.2.2 Troubleshooting GPA Current Sensors

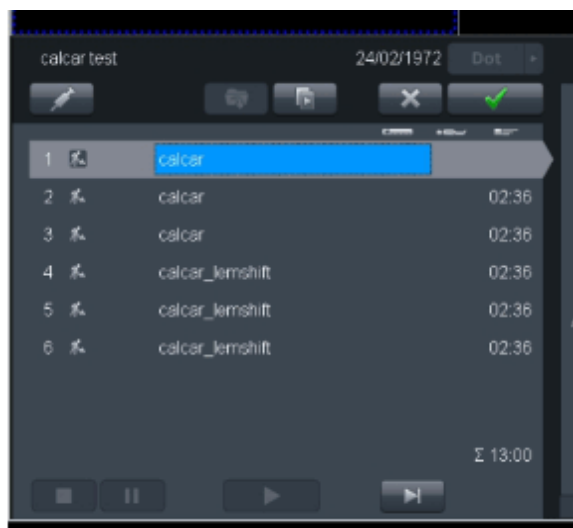
This document describes:

- How to find out defective (nonlinear) GPA current sensors.

2.2.2.1 Prerequisites

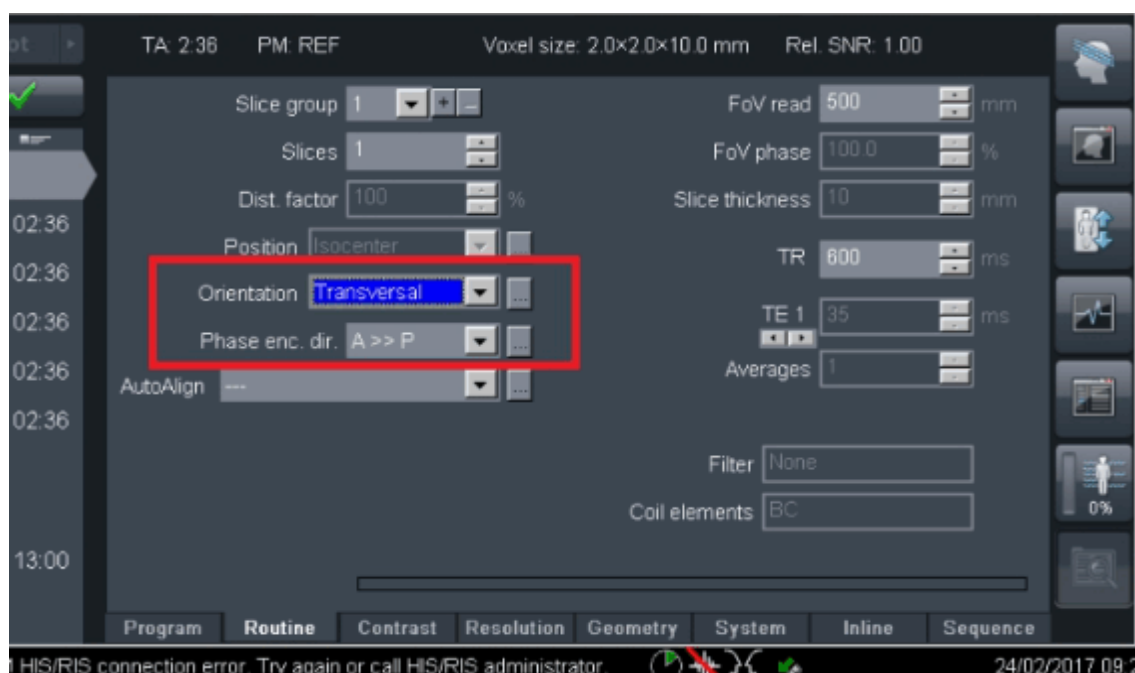
- Register a service patient
- Position a 240 mm spherical phantom (and loader for Tim systems)
- Navigate to the Siemens service sequences and copy 3 times calcar and 3 times calcar_lemshift

Fig. 2: Calcar test



- Calcar 1 – Orientation Transversal – Phase enc. Dir -> A-P
- Calcar 2 – Orientation Transversal – Phase enc. Dir -> R-L
- Calcar 3 – Orientation Coronal – Phase enc. Dir -> F-H
- Calcar_lemshift 1 – Orientation Transversal – Phase enc. Dir -> A-P
- Calcar_lemshift 2 – Orientation Transversal – Phase enc. Dir -> R-L
- Calcar_lemshift 3 – Orientation Coronal – Phase enc. Dir -> F-H

Fig. 3: Calcar test setup



- Run the 6 sequences in a row
- Open the viewing task card with a 3x3 layout
- Load the images in such a way:

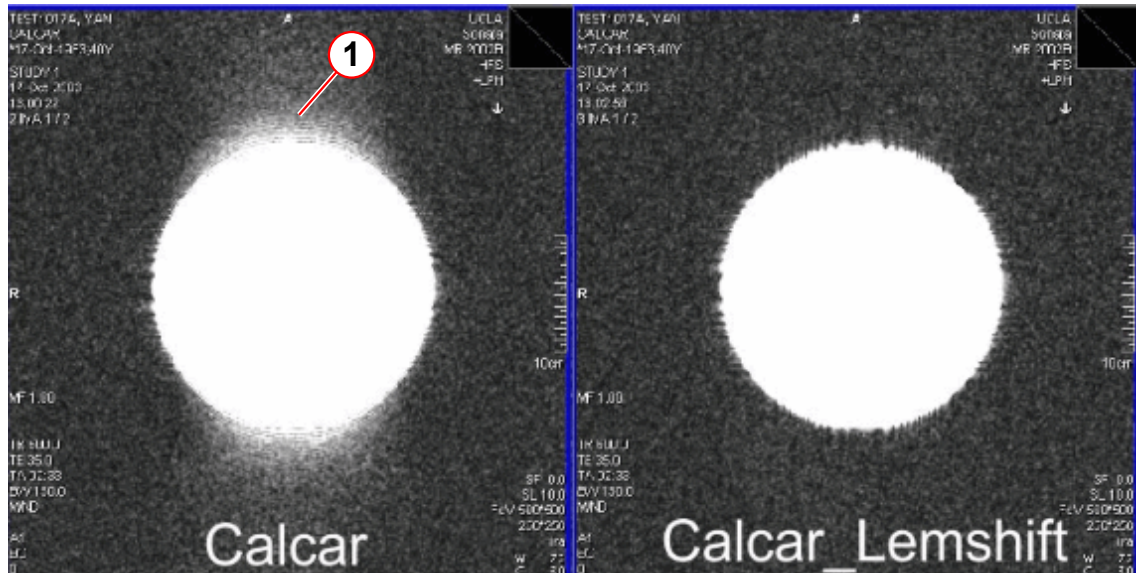
Calcar 1 (PE A-P) Y axis	Calcar 2 (PE R-L) X axis	Calcar 3 (PE F-H) Z Axis
Calcar_lemshift 1 (PE A-P) Y axis	Calcar_lemshift 2 (PE R-L) X axis	Calcar_lemshift 3 (PE F-H) Z axis
(empty)	(empty)	(empty)

- Compare the results of the two sequences for the same PE direction

- If calcar_lemshift is significantly better than calcar (less ghosting artifacts (→ Fig. 4 Page 13)), troubleshoot/ replace the corresponding current sensor of this axis.

Refer to REP Gradient.

Fig. 4: Calcar test - defective current sensor



(1) Ghosting Artifacts close to phantom

2.2.3 GPA Problems related to Implant Imaging Suite Option (Aera with FPO)

Aera systems with the Implant Imaging Suite option require a special hardware setup and configuration of the GPA cabinet. Otherwise the system may not work correctly, details are listed below.

Requirements:

D41 - Modulator assembly 10513415 Note: Rev. >=02 required for Implant Imaging Suite Option

D68 - Opto Link 10513414 Note: Rev. >=06 required for Implant Imaging Suite Option

D70 - I/O board 7563005 Note: Rev. >=02 required for Implant Imaging Suite Option

Configuration:

Check for correct jumper and switch settings of the GSSU. For details, see **CB-DOC / Service / System / Diagrams**.

2.3 Gradient Supervision (VLM)

This document describes:

- How to troubleshoot VLM problems
- How to install and use the VLM diagnostic Software "Siemens Medical VSC"
- The mandatory checks after a VLM Fire Alarm before handing over the system to the customer



Useful additional VLM information can be found under:

CB-DOC > Installation > Installation_Instructions > Gradient_Supervision and
CB-DOC > Installation > Startup > System Start-up > Startup of the Gradient Supervision.

2.3.1 General Troubleshooting VLM

2.3.1.1 Safety Information



WARNING

High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

» **Adhere to the described steps for your own safety.**

2.3.1.2 Prerequisites

For individual troubleshooting steps, **additional items** may be required:

- Standard Windows PC (WIN7 is compatible)
- VLM diagnostic Software "Siemens Medical VSC" (to be downloaded from CS-Intranet MR-product page)
- RS232 cable 9p-SUB-D female/male (material number 10915402)
- only if PC has no RS232 port:
Adapter USB - RS232 (material number 10861204)
- Measurement device for ambient atmospheric pressure differences (to be sourced locally, e.g. supplier of air condition systems)
- T-Distributor to be additionally installed at VLM aspirating hose system (material number 10683922)
- Adapter_D22_D25 bushing for VLM hose connection (material number 10683929)
- Temperature indicators (clock stickers) for gradient cables (material number 7071145)

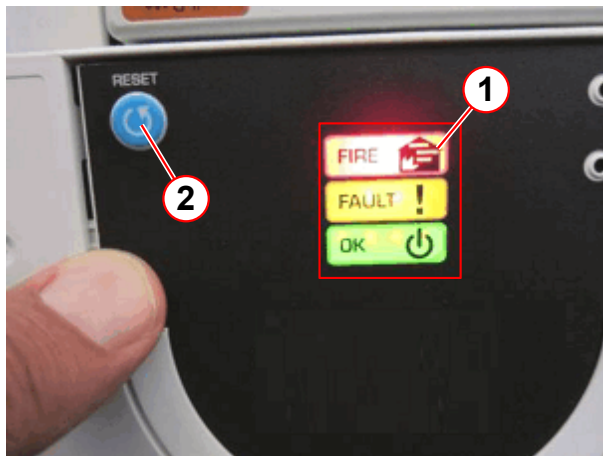
2.3.1.3 LEDs + Reset Button

Fig. 5: VLM front view



- The VLM-device has 3 status LEDs.
- A blue reset button is accessible when the front cover is opened.

Fig. 6: VLM LEDs + Reset Button



- (1) LEDs
(2) Reset Button

- GREEN = OK (default)
- YELLOW = FAULT (flashing, or static depending on the fault severity).
For SW versions >VB13, a customer pop-up is generated additionally.
- RED = FIRE (turns off the GPA modulator immediately and switches system to stand-by 1 min. later)
- The RESET-button is used for resetting FAULTs or normalizing the device (i.e., calibrating the 100% air flow value) .



The VLM device does only work properly under the following conditions:

- The air-flow through the VLM is between 12 and 24 l/min (allowed range for successful VLM normalization/calibration).
 - » Outside this range, the VLM device will show a warning (yellow FAULT LED on) and cannot be normalized/calibrated.
- The current air flow may deviate for max. +/- 30% from the normalized (calibrated) 100% value (RF-cabin door open/ closed).
 - » Deviations above this +/- 30% limit will lead to a FAULT condition of the VLM device and display of a customer pop-up.
 - » Deviations above +/- 20% from the normalized (calibrated) value will already lead to a "high air flow" fault but no customer alarm



At newer VLM devices with serial number $\geq$ B8712498, the maximum allowed deviation from the normalized (calibrated 100% air flow value) is $\pm 30\%$.
Older VLM devices ($<$ B8712498) have a smaller tolerance band of $\pm 20\%$.

2.3.2 Special Problem - Incorrect Air Flow

2.3.2.1 Problem Description + Troubleshooting Flowchart

The major problem currently seen in the field is an incorrect air flow through the VLM device.

There are several possible causes for that fault, see next section.



In case of incorrect air flow, the **VLM yellow FAULT LED is flashing or permanently ON.**

Fig. 7: Yellow Fault LED



Air-pressure related VLM problems Troubleshooting Flowchart

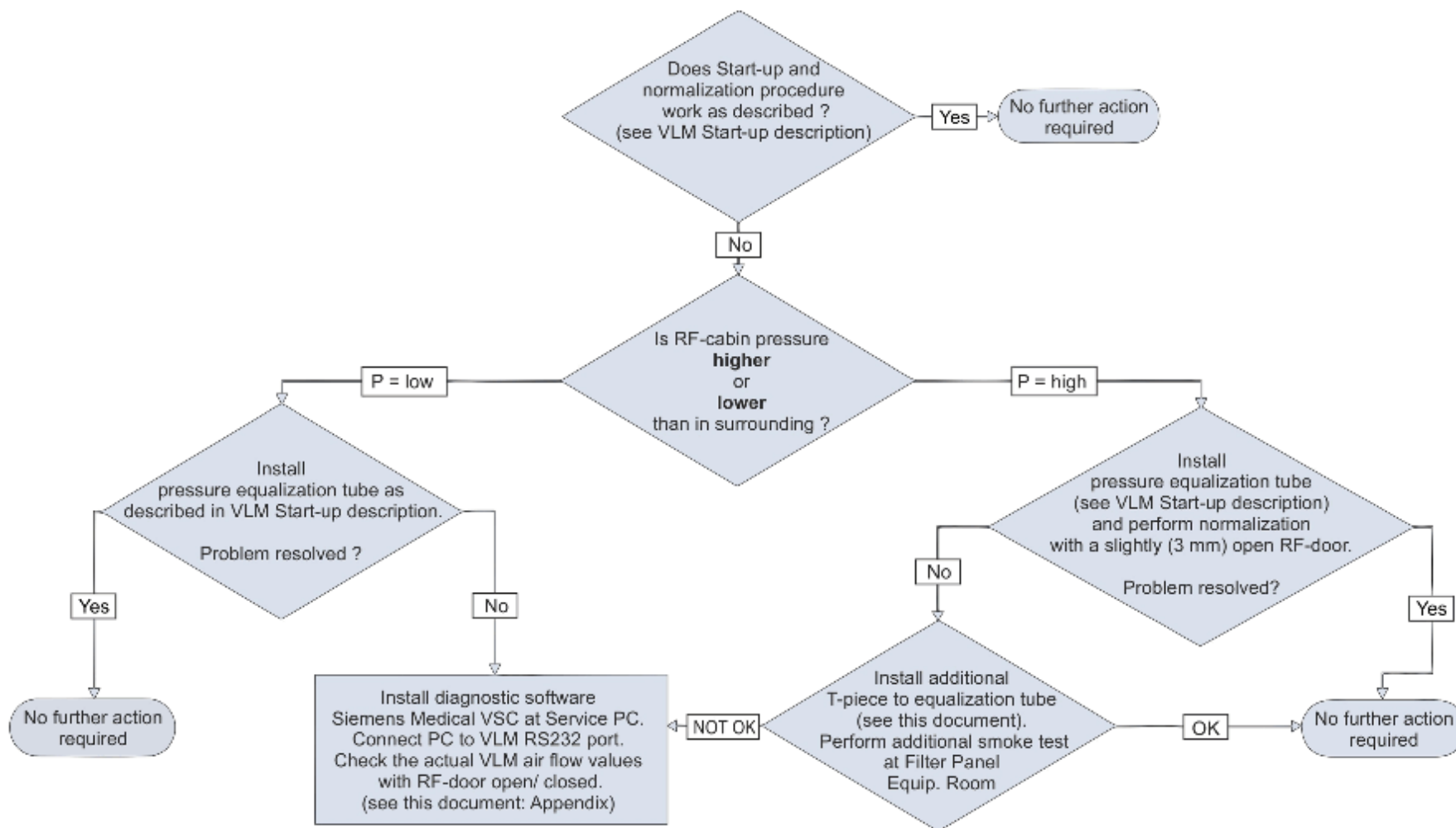
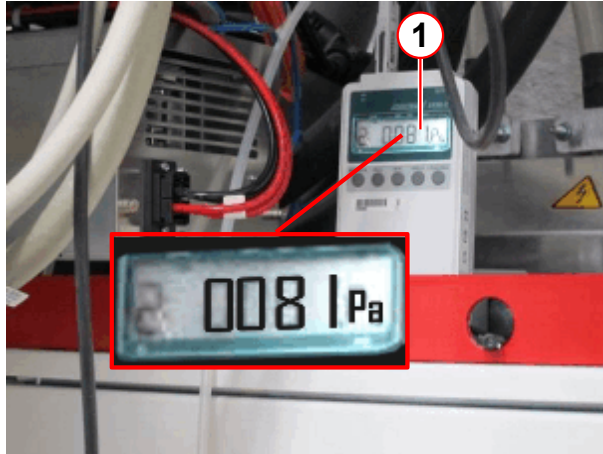


Fig. 8: Flowchart - VLM air pressure problems

2.3.2.2 Corrective Actions

Excessive Pressure Difference between RF-Cabin / Equipment Room

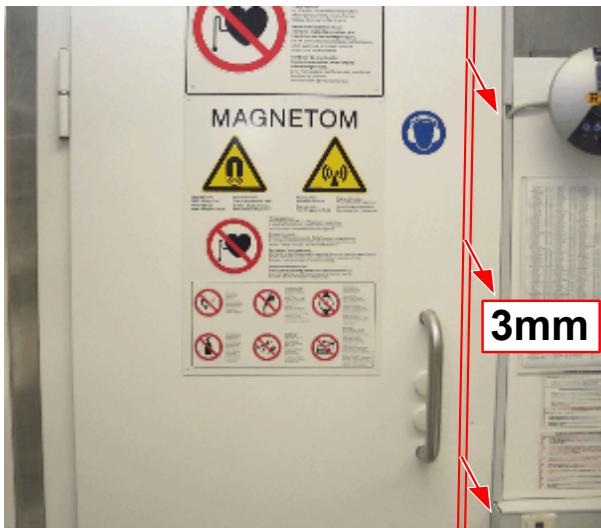
Fig. 9: Differential Pressure Measurement



(1) Example - Excessive diff. pressure > 81 Pa.

- The example on the left shows a site which was reporting VLM air flow problems.
- With help of a special measurement device (e.g. supplied by the air condition company), a differential pressure of 81 Pascal was measured between RF-cabin and equipment room!
- For a fault-free operation of the VLM, the required value is <50 Pa (0.5 mBar).
 - » The optimum would be, to have **NO differential pressure** at all!
 - » The air conditioning system has to be corrected (e.g., installation of additional exhaust air duct).

Fig. 10: Open RF door 3 mm



VLM normalization (calibration) with slightly open RF door:

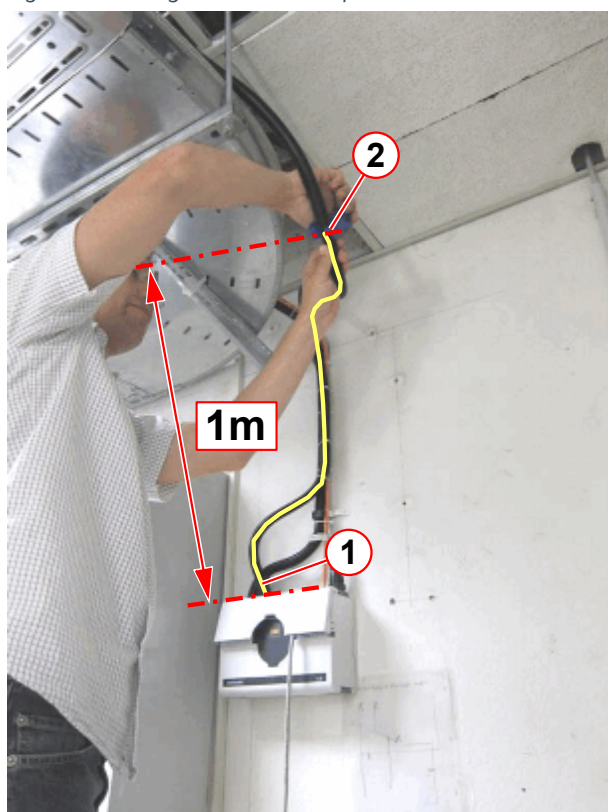
- A slightly modified VLM normalization procedure (calibration) of the 100% air flow value (with slightly open RF-door) can help to improve the situation. This is only applicable in case of a moderate pressure difference between RF-cabin and equipment room.
- The VLM device is powered up and operating for at least 90 sec. warm-up time.
- Open the RF cabin door slightly (~3 mm).
- Open the front cover of the device and push the RESET button for 10 sec. (→ Pos.2/Fig. 6 Page 15) until all LEDs are flashing.
- Release the RESET button and wait until the LEDs go off.
- When the LEDs start flashing again, press the RESET button once more to confirm the actual calibration.
 - » This results in a normalization at a mean cabin air pressure between maximum (door closed) and minimum (door open) and allows a maximum of tolerance from normalized (calibrated) value in both directions.



The diagnostic Software Siemens Medical VSC can be used to measure the VLM air flow and determine the optimum opening gap of the door for normalization.

See (→ Appendix - Diagnostic Software (Siemens Medical VSC) / Page 34).

Fig. 11: Cutting the Pressure Equalization Hose



- (1) VLM outlet
- (2) Cut hose 1 m after the VLM outlet

If the previous step did not help, install additional T-piece:

- An additional T-piece installed at the pressure equalization hose may help.
- Cut the pressure equalization hose about 1 m behind the VLM outlet (see left figure (2)).

Fig. 12: T-piece with 2 m hose



- Prepare a T-piece with approx. 2 m of hose as shown on the left.
- The T-piece can be ordered as a standard spare part with material number 10683922 (this T-piece has an additional reduction to 10 mm fitted, which must be removed).
- If needed, the black bushing at the VLM-in/ -outlet can be also ordered as a spare part (Adapter_D22_D25, material number 10683929).

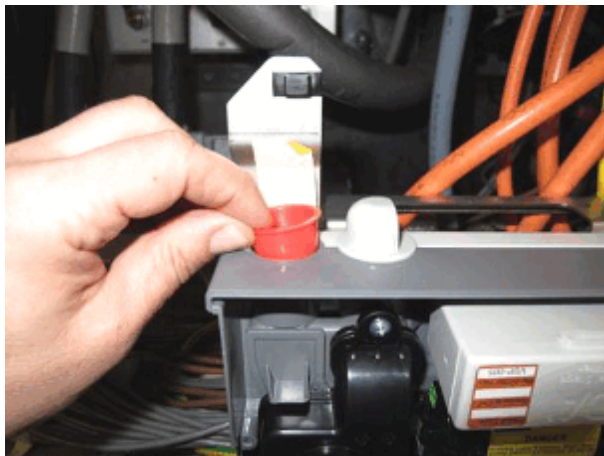
Fig. 13: T-piece connected



- Connect the T-piece to the previously cut ends of the pressure equalization hose as shown on the left.
- » The outlet of the 2 m hose must remain inside the equipment room.

VLM Blind Cap not removed

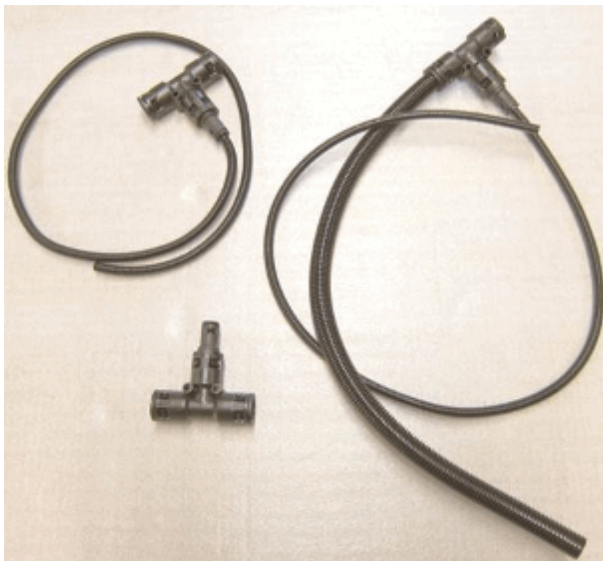
Fig. 14: Remove Blind Cap



- Here, a red blind cap was forgotten at the VLM input.
- This will result in zero flow and a FAULT condition.
- » Remove blind caps from the VLM before installing the aspirating hose system.

Leaking Hose Connections

Fig. 15: VLM Hose Connections



The aspirating hose system may leak in case:

- a tube is not completely pushed into it`s manifold (T-piece...). Check turning moment between manifold and tube:
 - » Good fitting = noticeable resistance.
 - » Bad fitting = no resistance for turning the tube.
- a corrugated tube is not properly cut and does not have smooth edges.
 - » Cut the tube correctly with the delivered tool.
 - » See VLM Installation Instructions: **CB-DOC>Installation>Installation Instructions>Gradient Supervision**

Fig. 16: Proper cut

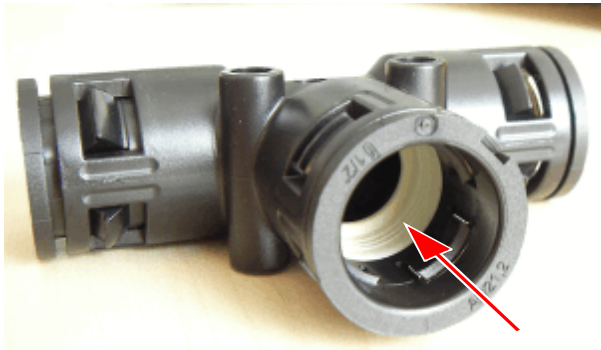


Fig. 17: Improper cut



Missing Joint Seals

Fig. 18: Joint Seal (white)

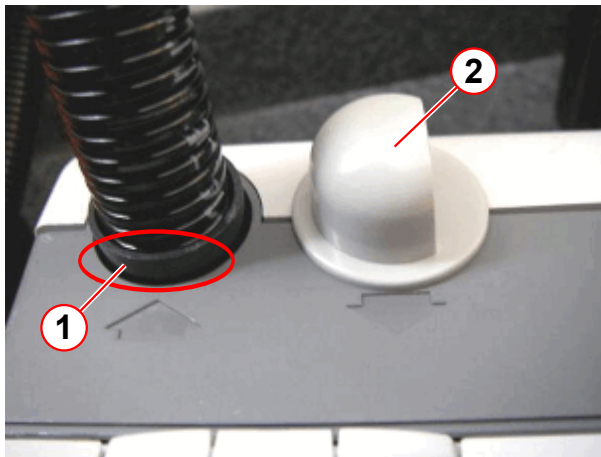


The white rubber joint seal must be properly fitted:

- In case of a FAULT condition, check the joints (T-pieces) for proper sealing.
- A missing seal easily results in an air flow difference of 1.4 l/min. (~ 7%).
 - » Check if the white rubber seal is fitted in each joint as shown on the left.
 - » In case the seal is missing, order a new T-piece (material number 10683922)

Missing Bushing

Fig. 19: VLM Hose Connection



- (1) Black Bushing
(2) Grey Outlet Cover

The black bushing must be always used when a hose is connected to the VLM inlet or outlet:

- If the black bushing (1) (Adapter_D22_D25, material number 10683929) is missing, the flow will increase dramatically for 9 l/min. (~ 43%).
- When the grey outlet cover (2) is removed, the flow increases for 0.4 l/min. (~ 2%)
 - » Check if the black bushing is fitted to the VLM inlet and, if applicable, to the VLM outlet hose connection.
 - » Check if the grey outlet cover is fitted to the VLM outlet (only in case no return hose is fitted).

2.3.3 Special Problem - Incorrect VLM Configuration

2.3.3.1 Problem Description



In case the VLM configuration jumper is set incorrectly, **the VLM will show NO fault indications, but the GPA will be permanently disabled with an error message.**

The VLM has a configuration jumper which must be set according to the system type.

- This jumper is inside the VLM..
- It defines the logic of the Modulator_off signal which is forwarded via a fiber-optic cable to the GSSU D70 I/O board, U1.
- In normal condition (NO fault), the light at FOC U1 must be ON.
- In case of a VLM FIRE Alarm (at Tim systems also in case of Gradient Coil overtemperature):
 - the light at U1 will be OFF
 - the GPA modulator will be switched off immediately
 - a GPA error message is generated.

2.3.3.2 Corrective Action

Check VLM Jumper Settings

Fig. 20: Jumper on 2-3 default setting - correct for Tim systems and Biograph mMR

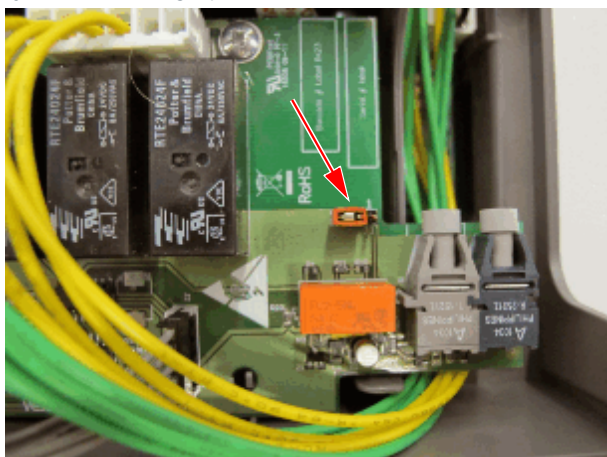
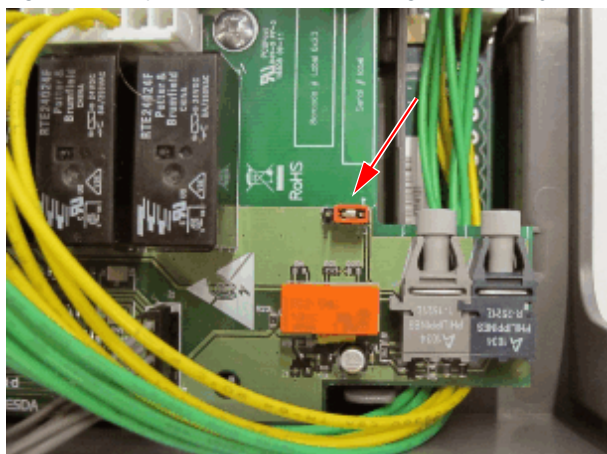


Fig. 21: Jumper on 1-2, correct setting for Aera/Skyra)

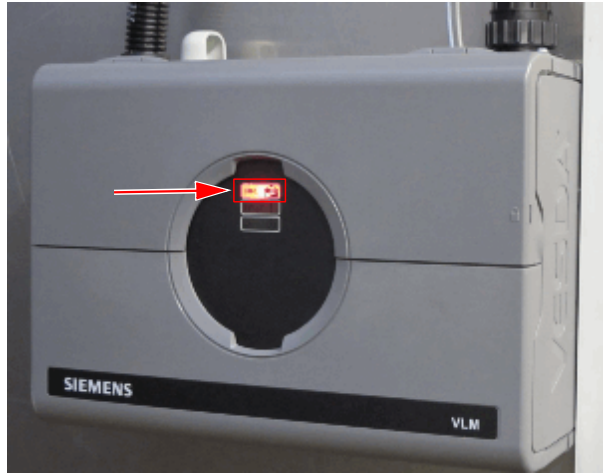


- Open the VLM cover
- Inside, there is one jumper left of the fiber optic connectors:
 - » The default jumper setting is 2-3, which is appropriate for all **Tim systems, Terra, and Biograph_mMR.**
 - » On **Aera/ Skyra/ Prisma/ Avanto^{fit}/ Skyra^{fit} / Prisma^{fit}** systems, this jumper must be set to 1-2 as per (→ Fig. 21 Page 24)

2.3.4 System Inspection after VLM FIRE Alarm

After a VLM FIRE Alarm was triggered for unknown reason, the following work steps must be successfully performed before handing over the system to the customer:

Fig. 22: VLM FIRE Alarm



2.3.4.1 Checklist



All system inspection steps must be successfully performed and checked with OK before handing over the system to the customer!

Tab. 1 System Inspection VLM FIRE Alarm - Checklist

All checks must be OK before returning the system back to the customer.			
Work Step	OK	Not OK	Procedure
MR system is powered-off WARNING: High voltage! See Safety Information.	☐	n.a.	(→ Prerequisites / Page 29)

All checks must be OK before returning the system back to the customer.			
Work Step	OK	Not OK	Procedure
NO signs of excessive heat around VLM Sample Points			(→ Check around Air Sample Points / Page 29)
Magnet - Lower Sample Plate: - Gradient Coax Cables - GC connections - Body Coil			
Magnet - Upper Sample Plate: - Gradient Coax Cables - Gradient Cables to RF Filter Panel - Gradient connections, bolts, nuts, washers			
RF-Filter Panel - Examination Room: - Gradient Cables - Bolts, nuts, washers - Gradient Filters	☐	☐	
RF-Filter Panel - equipment room: - Gradient Cables - Bolts, nuts, washers - Gradient Filters			
Inside EPC/ACC and GPA cabinet: - EPC/GPA Transformer - GPA Chokes - Other components			

All checks must be OK before returning the system back to the customer.			
Work Step	OK	Not OK	Procedure
NO External Causes found - NO craftsmen work in vicinity of site (welding, dust, etc.) - NO people smoking - NO use of fossil flammable material in the surrounding (heating systems) - NO air condition defects or distribution of external pollution through the air ducts (specially check inlet) - NO excessive environmental pollution	☐	☐	(→ Exclusion of External Causes / Page 29)
VLM Event Log - Download, analysis of Event Log - Event Log sent to RSC/ HSC	☐	☐	(→ Analysis of VLM Event Log / Page 30)
VLM Reset - FIRE Alarm + FAULT LED off	☐	☐	(→ Reset of VLM / Page 31)

All checks must be OK before returning the system back to the customer.			
Work Step	OK	Not OK	Procedure
Final Checks			(⇒ Final Checks / Page 31)
- System switched-off			
- NO activated temperature indicators (clock stickers at gradient cables) before system switch-on.			
WARNING: High voltage! See Safety Information.			
- System switch-on			
- NO increased smoke background level during running of service sequence "gradfree-pulse"			
- NO activated temperature indicators (clock stickers at gradient cables) after running "gradfreepulse"	☐	☐	
WARNING: High voltage! See Safety Information.			
- NO spikes found with QA>Spike Check			
- Check of Image Orientation QA>Image Orientation Check (only if gradient cables were removed, a new measurement must be performed)			

2.3.4.2 Prerequisites



WARNING

High voltage

Contact with voltage-conducting parts will lead to death or serious physical injury.

» Adhere to the described steps for your own safety.

- Ensure, the system is switched off
- Switch off circuit breakers:
 - for Aera/ Skyra/ Prisma/ Prisma^{fit}: **F21/EPC and F1/GPA**
 - for Avanto, Avanto^{fit}, Skyra^{fit}, Biograph_mMR, Verio, Terra, and Trio A Tim System : **F2/GPA/GSSU and F1/GPA**
 - for Symphony A Tim System: **F6/GSSU in the ACC cabinet and F110/GPA at the mains box**
- Wait at least for 20 minutes to ensure the final stage capacitors are fully discharged.

2.3.4.3 Check around Air Sample Points

Check for any signs of excessive heat in the vicinity of all air sample points:

- At magnet lower sample plate (GC Coaxial Cable Connections)
- At magnet upper sample plate (Gradient Cable Connections, bolts, screws, etc.)
- At RF-filter plate Examination Room (Gradient Cable Connections, Gradient Filters)
- At RF-filter plate equipment room (Gradient Cable Connections, Gradient Filters)
- Inside EPC/ACC and GPA cabinet (System- and GPA-transformer, GPA-chokes, other electronic components)



Overheated components/ connections might be recognized by the following signs:

- Discoloration, color change (e.g. copper conductors)
- Melted insulation (cables)
- Black marks from sparks, smoke residuals
- Activated temperature indicators (clock stickers at gradient cables)
- Bad smell

2.3.4.4 Exclusion of External Causes

Exclude external causes (particle release) which might have triggered a VLM FIRE Alarm, e.g.:

- Craftsmen work (welding, dust, etc.)
- People smoking
- Use of fossil flammable material in the surrounding (heating systems)

- Air condition defects or distribution of external pollution through the air ducts (specially check inlet)
- Excessive environmental pollution



Check smoke graph in diagnostic Software "Siemens Medical VSC"
(→ Pos.2/Fig. 36 Page 38).

For countries with strong environmental pollution, a VLM-device with higher alarm threshold is available (material number 109 15 504, HSC release required).

2.3.4.5 Analysis of VLM Event Log

- Evaluate the VLM Event Log, see (→ Appendix - Diagnostic Software (Siemens Medical VSC) / Page 34).
- Save the VLM Event Log as file (→ VLM Event Log / Page 41) and send it to RSC/ HSC for further analysis.



Check time stamps of FIRE Alarm in the Event Log (→ VLM Event Log / Page 41).

If a FIRE Alarm was reported at times where the MR system was powered off, gradient connections and GPA are not the problem cause.

In this case, investigate External Causes (→ Exclusion of External Causes / Page 29) and within EPC/ACC cabinet.

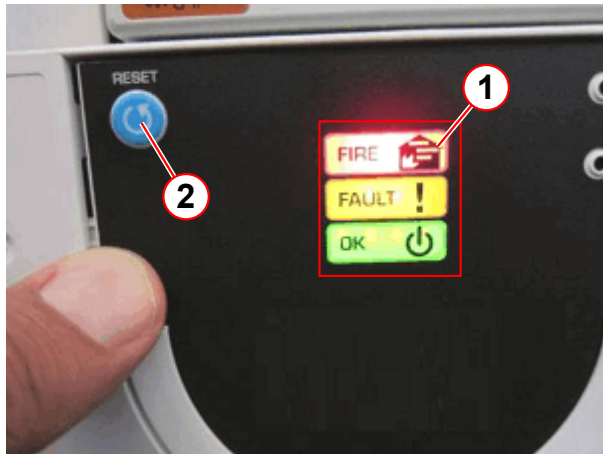
2.3.4.6 Reset of VLM

Fig. 23: VLM front view



- Open the VLM front cover.

Fig. 24: VLM LEDs + Reset Button



- (1) LEDs
- (2) Reset Button

- Push the blue RESET-button (2).
 - » The red FIRE LED (1) should be extinguished.

2.3.4.7 Final Checks

- Prior to running measurements, check the condition of all temperature indicators (clock stickers) at the gradient cables.



CAUTION

High voltage!

Touching any of the gradient connections may result in serious injury or death.

- » Do not touch any of the gradient connections, or power down the system according to (→ Prerequisites / Page 29), prior to the following work steps.

Fig. 25: Old temperature indicator (clock sticker)_NOT activated

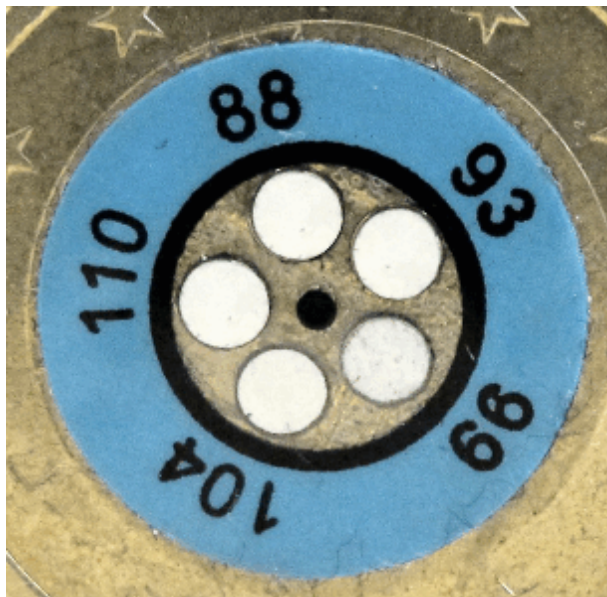
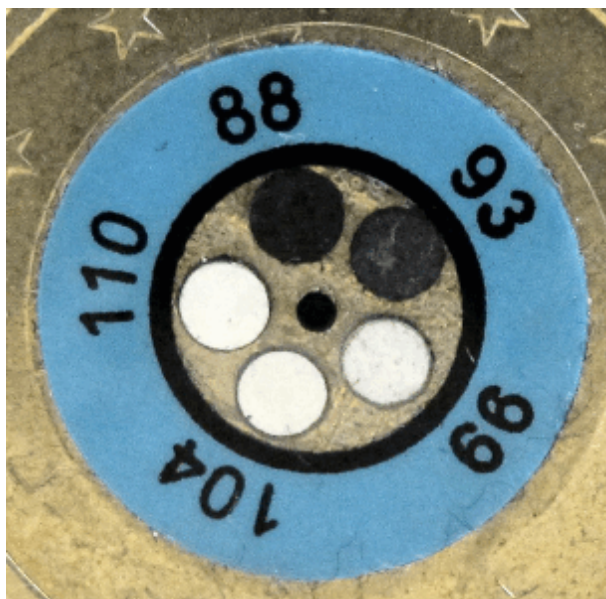


Fig. 26: Old temperature indicator (clock sticker)_Activated



- Visually check all of the temperature indicators (clock stickers) for integrity.

- » None of the labels must show a change in color!

- An **activated temperature indicator** is a sign for a poor electric contact and high resistance due to loose or bad contacts.

- » Check/ correct the connection in doubt.
- » EXAMPLE - the left figure shows an activated temperature indicator, which has been exposed to $>93^{\circ}\text{C}$ and $<99^{\circ}\text{C}$.



In case of missing or damaged clock stickers, attach new temperature indicators (material number 7071145). Details see UI MR029/10/S.

- Switch on circuit breakers:
 - for Aera/ Skyra/ Prisma/ Prisma^{fit}: **F1/GPA and F21/EPC**
 - for Avanto, Avanto^{fit}, Skyra^{fit}, Biograph_mMR, Verio, Terra, and Trio A Tim System : **F1/GPA and F2/GPA/GSSU**
 - for Symphony A Tim System: **First, F110/GPA at the mains box and then F6/GSSU in the ACC cabinet.** (Use this order to avoid damage!)
- Switch the MR system "ON" at the Alarm Box.

- Go to Service **Software>TestTools>Tools**.
- Register a Service Patient.
- Set adjustments to "Service Mode".
- Load service sequence "gradfreepulse".



This check is done to prove the integrity of the critical gradient connections at high load.

- Set the sequence parameters according to the following table:

Tab. 2 Sequence parameters to be set

Parameter	Value	Unit	Definitions
TI 1	10	mT/m	Gradient strength
TI 3	10	mT/m	Gradient strength
TI 5	10	mT/m	Gradient strength
TI 2	3	n.a.	
TI 4	3	n.a.	
TI 6	3	n.a.	
TI 7	1	n.a.	
TD 1	0.1	ms	Rise time
TD 3	0.1	ms	Rise time
TD 5	0.1	ms	Rise time
TD 2	0.3	ms	Flat top
TD 4	0.3	ms	Flat top
TD 6	0.3	ms	Flat top

- Set averages and number of measurements to achieve a time of ~28 min.
 - » Other parameters must not be changed.
- Run sequence while observing smoke graph with VLM diagnostic Software (→ Pos.2/Fig. 36 Page 38).
 - » The smoke background level must not increase during measurement!
Otherwise, check/ repair overheated gradient connections or other overheating parts (e.g. Body Coil).

- After measurements, check again the condition of the **temperature indicators**.

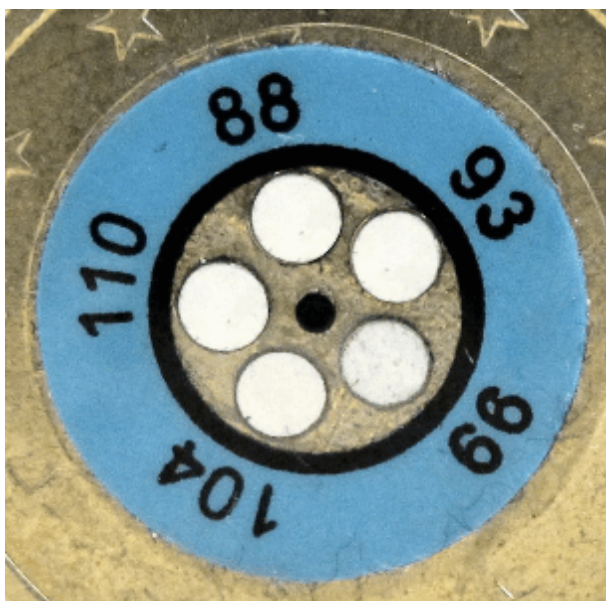
! CAUTION

High voltage!

Touching any of the gradient connections may result in serious injury or death.

- » Do not touch any of the gradient connections, or power down the system according to (→ Prerequisites / Page 29), prior to the following work steps.

Fig. 27: Old temperature indicator (clock sticker)_NOT activated



- Visually check all of the temperature indicators (clock stickers) for integrity.
 - » None of the labels must show a change in color!

- Perform **QA>Spike Check** at least three times..
 - » If spikes were found, check the gradient connections again. If required, follow the **CB-DOC>TSG>TuneUp/QA>Spike Troubleshooting**
- If any of the gradient connections have been removed, **QA>Image Orientation Check** must be performed.

2.3.5 Appendix - Diagnostic Software (Siemens Medical VSC)

2.3.5.1 Overview

For a detailed troubleshooting of the VLM, the diagnostic Software "Medical VSC" is provided which can be installed at the PC.

2.3.5.2 Requirements

- Standard Windows PC (WIN7 is compatible)
- RS232 cable 9p-SUB-D female/male (material number 10915402)

- only if PC has no RS232 port:
Adapter USB - RS232 (material number 10861204)



In case your PC has no RS232 port:

- Use adapter USB - RS232 (material number 10861204). Other available standard adapters are usually not compatible with the VLM-device and no communication is possible.
- Installation of a driver might be required at the PC. Follow instructions printed at the packing of the USB - RS232 adapter.

2.3.5.3 Software Download

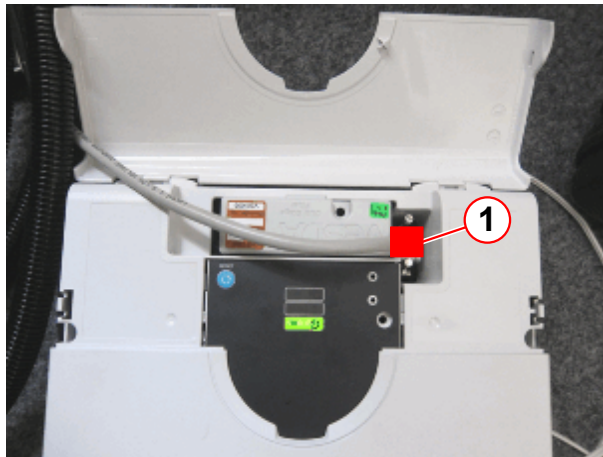
Fig. 28: Icon - Siemens Medical VSC



- Download the VLM Software_Win7.ZIP file from the Intranet **Product Information>MR>Downloads>Gradient_Supervision**
- Unzip it to your PC and install the SiemensMedicalSetup_30500.msi
- You will get an icon at your desktop:

2.3.5.4 Establishing Communication to VLM

Fig. 29: VLM - open top cover



(1) RS232 connector

- Connect the RS232 cable (male connector (1)) to the socket on front of the VLM-device (under the top cover).
- Connect the other end of the cable to your PC.

Fig. 30: Icon - Siemens Medical VSC



- At the PC, double-click the desktop icon:

- The VLM user interface will be shown (see below).

Fig. 31: VLM User Interface

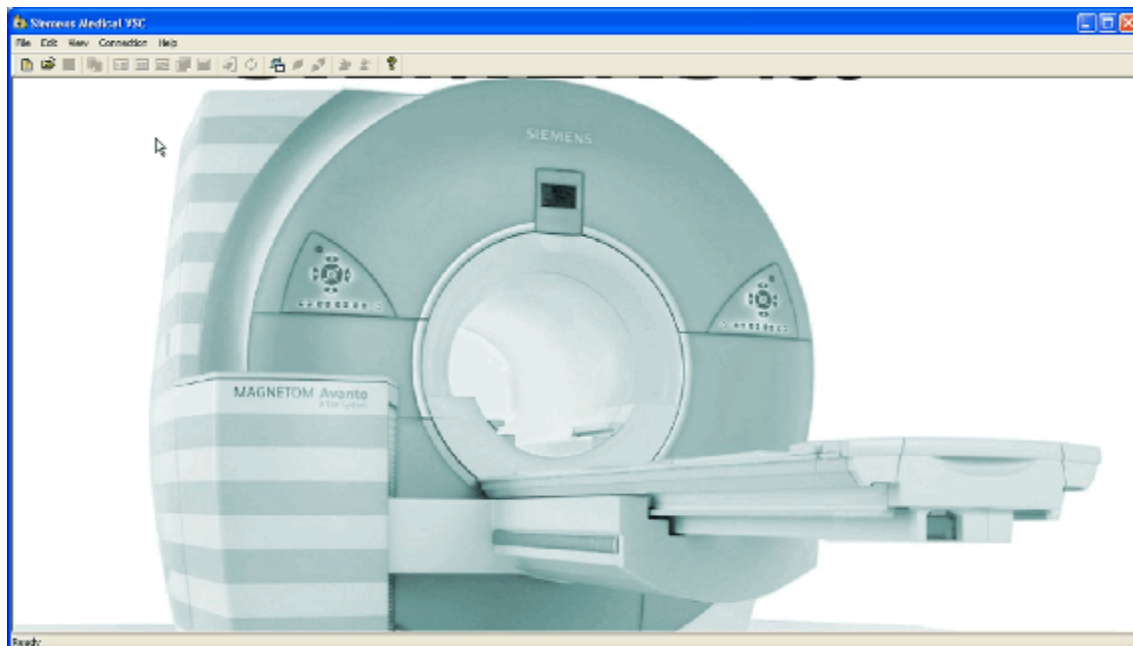
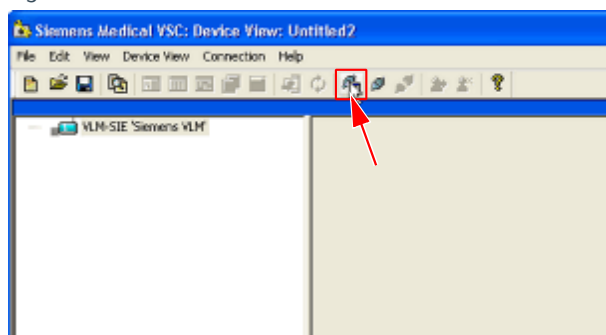
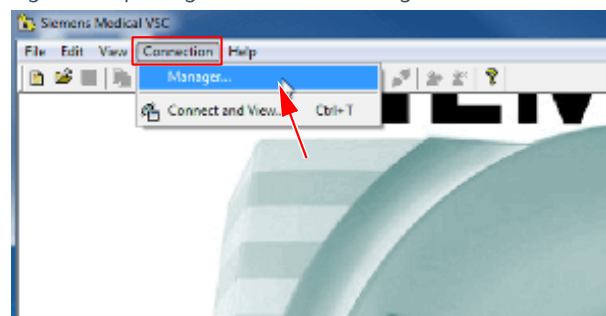


Fig. 32: "Connect" button



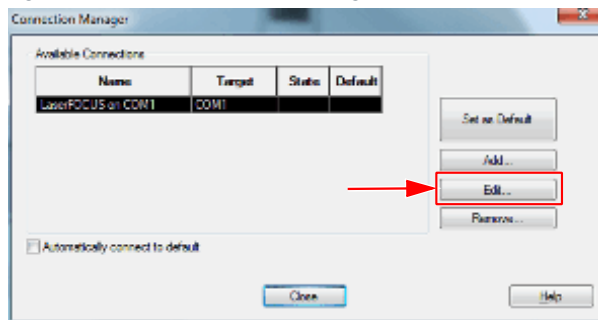
- Click the **"Connect"** button in order to start communication with the VLM-device.

Fig. 33: Opening "Connection Manager"



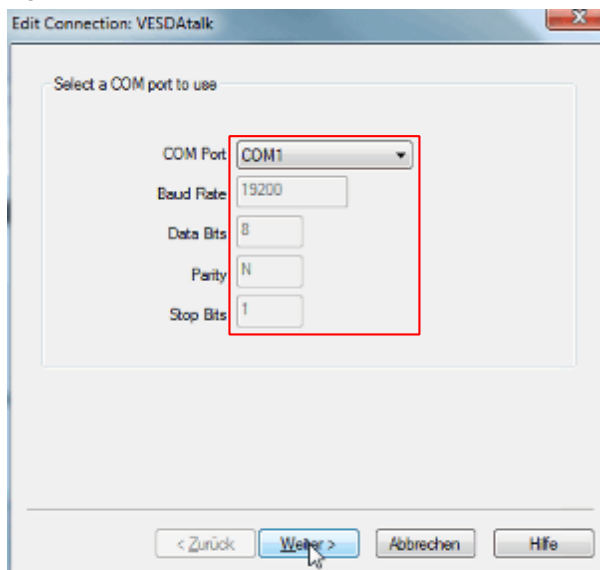
- If the communication parameters are not set properly, the communication will fail. In this case please open the Connection Manager.

Fig. 34: Edit - Connection Manager



- Click on "Edit"...

Fig. 35: Communication Parameters

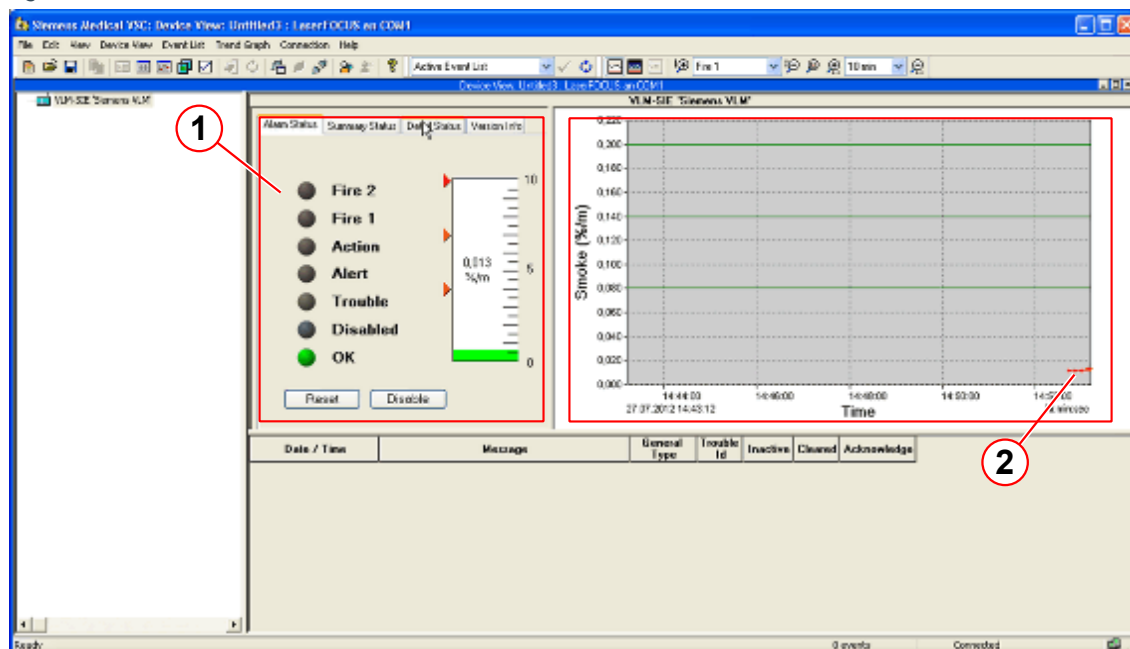


- ...and check/ correct the communication parameters according to the left figure.
 - COM Port used (here: COM1)
 - Baud Rate = 19200
 - Data Bits = 8
 - Parity = N
 - Stop Bits = 1

2.3.5.5 VLM User Interface

- After establishing communication to the VLM, the Device View will open with the Alarm Status (1) and Smoke Graph (2).

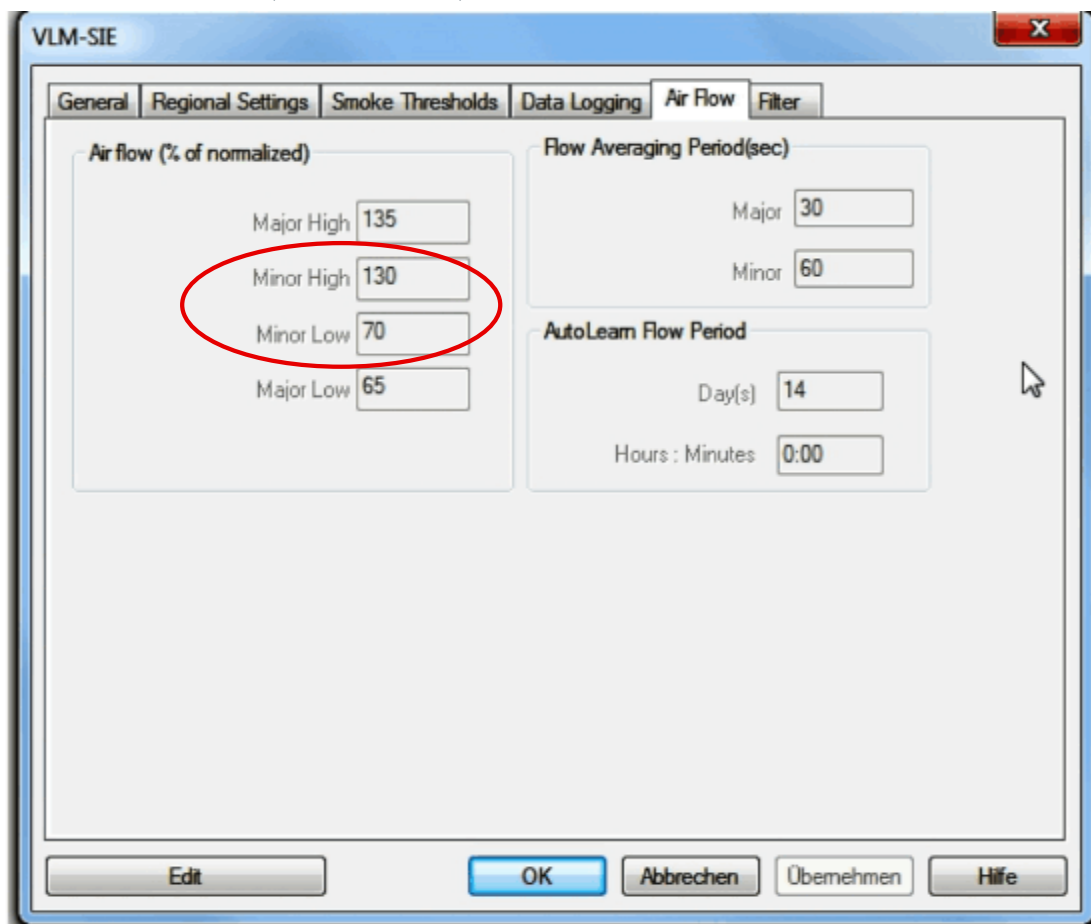
Fig. 36: Device View



- (1) Alarm Status
(2) Smoke Graph

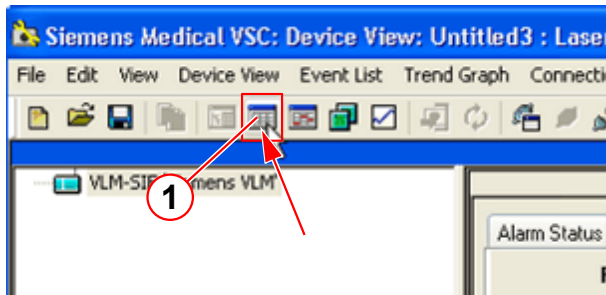
- Click the "EDIT" button, and select "EDIT" in the submenu to check the programmed air flow threshold.

Fig. 38: Air flow thresholds (VLM $\geq$ B8712498)



2.3.5.7 VLM Event Log

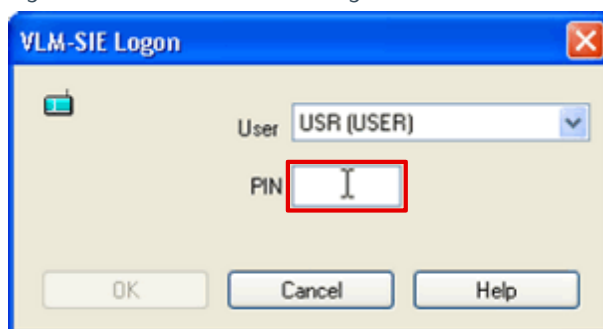
Fig. 39: Starting VLM Event Log



(1) Event Log button

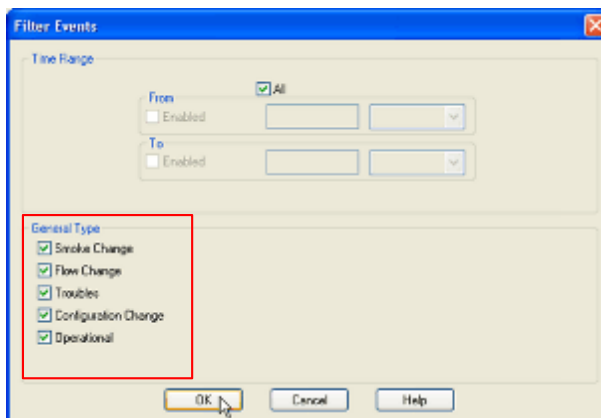
- Click on the “**Event Log**” button (1) to see the list of logged events.
- The list can be also downloaded with **File>Save as**.

Fig. 40: Password VLM Event Log



- The Software will ask you for a 4-digit PIN to get access to the Event Log.
- Enter **1111** and click OK.

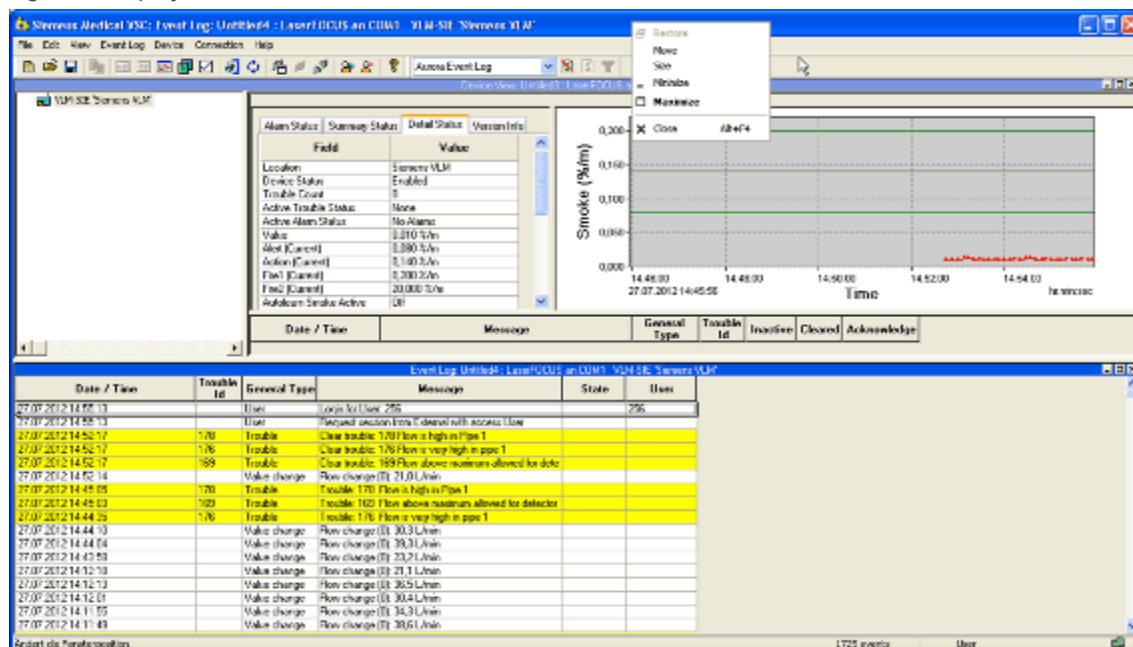
Fig. 41: Filter Settings



- By default, all filters are selected.
- Click “OK” to confirm configurations.

- Now you see the Event Log.
 - » Yellow marked events are FAULT conditions.
 - » Red events are FIRE conditions.
 - » The time stamps (VLM Device Time) might differ from the MR system time.

Fig. 42: Display of Events



There might be an offset between the Event Log time (= VLM Device Time) and the site-specific MR system time:

- The standalone battery-buffered **VLM clock cannot be set by the CSE**.
- The **VLM clock is set to UTC** (Universal Time, Coordinated) by default.
- Example: To correct the time stamp in the error log of the VLM device from UTC to e.g. CET (Central European Time), add +1 hr. Or, in case of daylight saving time CET DST, +2 hrs. must be added.

Version 02:

- New system variants MAGNETOM **Avanto^{fit}** and **Skyra^{fit}** added.

Version 03:

- New chapter (→ Procedures / Page 9) added - procedure to isolate GPA rectifier defects included (→ D100 Rectifier - Verifying a defect / Page 10).

Version 04:

- Chapter (→ Procedures / Page 9) - new section (→ Gradient Supervision (VLM) / Page 14) added.

Version 05

- Chapter (→ Procedures / Page 9) (→ Gradient Supervision (VLM) / Page 14) updated, air flow threshold +/- 30% up from serial number B8712498, Charm MR_00464686.

Version 06

- Chapter (→ Procedures / Page 9) (→ Troubleshooting GPA Current Sensors / Page 11) added.

Version 07

- Chapter (→ Procedures / Page 9) (→ GPA Problems related to Implant Imaging Suite Option (Aera with FPO) / Page 13) added.

There are no Hazard IDs in this document.

- Restricted - All documents may only be used by authorized personnel for rendering services on Siemens Healthcare Products. Any document in electronic form may be printed once. Copy and distribution of electronic documents and hardcopies is prohibited. Offenders will be liable for damages. All other rights are reserved.

siemens-healthineers.com

Siemens Healthineers Headquarters
Siemens Healthcare GmbH
Henkestr. 127
91052 Erlangen
Germany
Telephone: +49 9131 84-0
siemens-healthineers.com

Print No.: M7-000.840.16.07.02 | Replaces: M7-000.840.16.06.02
Doc. Gen. Date: 07.18 | Language: English
© Siemens Healthineers, 2010